

CAPE CHARLES 5 ENE, VA, USA

WMO: 745900

Lat: 37.2907N

Lon: 75.9270W

Elev: 9

StdP: 101.22

Time Zone: -5.00 (NAE)

Period: 04-19

WBAN: 03739

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -6.7	(c) -4.6	(d) -16.5	(e) 0.9	(f) -6.4	(g) -13.7	(h) 1.1	(i) -4.0	(j) 10.0	(k) -1.2	(l) 8.8	(m) -1.6	(n) 3.9	(o) N/A	(p) 0.372

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 7.3	(c) 32.3	(d) 24.9	(e) 31.0	(f) 24.6	(g) 29.9	(h) 24.2	(i) 26.5	(j) 30.0	(k) 25.8	(l) 29.1	(m) 25.2	(n) 28.4	(o) 2.5	(p) N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
25.5	20.8	28.1	24.9	20.0	27.6	24.3	19.2	27.1	82.4	30.0	79.4	29.1	76.9	28.5	29.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 7.6	(b) 6.6	(c) 5.7	(e) -9.4	(f) 34.9	(g) 2.5	(h) 1.6	(i) -11.2	(j) 36.0	(k) -12.7	(l) 37.0	(m) -14.1	(n) 37.8	(o) -16.0	(p) 39.0		
(4) 7.6	(5) 6.6	(6) 5.7	(7) -9.4	(8) 34.9	(9) 2.5	(10) 1.6	(11) -11.2	(12) 36.0	(13) -12.7	(14) 37.0	(15) -14.1	(16) 37.8	(17) -16.0	(18) 39.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	15.1	4.1	5.0	8.2	13.9	18.9	23.4	25.7	24.8	22.1	16.6	10.5	6.8	(6)
(7)		DBStd	8.71	5.41	5.17	5.00	4.63	4.00	3.09	2.40	2.35	3.03	4.25	4.38	4.86	(7)
(8)		HDD10.0	636	196	156	97	14	0	0	0	0	2	47	125		(8)
(9)		HDD18.3	2008	443	373	314	148	43	2	0	6	84	237	357		(9)
(10)		CDD10.0	2483	12	16	43	130	277	402	487	458	363	208	61	26	(10)
(11)		CDD18.3	812	0	0	1	14	62	154	229	200	118	31	2	1	(11)
(12)		CDH23.3	6012	0	0	6	68	294	1127	2096	1642	669	108	1	1	(12)
(13)		CDH26.7	1559	0	0	0	8	54	307	648	417	114	11	0	0	(13)
(14)	Wind	WSAvg	2.5	2.9	3.0	3.2	2.9	2.4	2.1	1.9	1.9	2.2	2.3	2.6	2.5	(14)
(15)	Precipitation	PrecAvg	1156	80	75	103	86	85	98	124	121	117	98	75	93	(15)
(16)		PrecMax	1551	176	189	243	126	180	222	258	279	295	256	198	207	(16)
(17)		PrecMin	873	42	12	11	24	17	42	14	44	19	1	4	29	(17)
(18)		PrecStd	181	29	39	44	27	41	40	59	58	73	56	48	40	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	19.4	21.5	23.9	28.0	30.2	33.6	34.8	33.4	30.9	27.9	22.7	21.6	(19)
(20)			MCWB	16.6	17.2	17.7	20.8	22.7	24.4	25.5	25.5	24.4	22.5	18.8	19.0	(20)
(21)		2%	DB	16.6	18.3	20.8	25.4	28.2	31.5	32.6	31.4	29.5	25.9	20.4	18.9	(21)
(22)			MCWB	12.9	14.7	15.0	18.5	21.6	23.7	25.4	25.0	24.1	22.1	17.8	16.7	(22)
(23)		5%	DB	14.5	15.8	18.5	23.1	26.5	29.9	31.2	30.1	27.9	24.4	18.7	16.6	(23)
(24)			MCWB	11.3	12.5	14.9	18.5	21.3	23.3	25.1	24.6	23.7	21.7	16.3	14.7	(24)
(25)		10%	DB	12.3	12.9	16.2	20.8	24.9	28.4	30.1	29.0	26.7	22.9	17.1	14.3	(25)
(26)			MCWB	10.0	10.3	13.1	16.6	20.7	22.9	24.7	24.0	23.1	20.2	14.7	12.9	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	16.6	17.9	18.3	22.2	24.3	25.7	27.8	26.9	25.8	24.6	20.0	19.9	(27)
(28)			MCDWB	18.8	20.6	21.9	25.0	27.5	29.2	30.9	29.9	28.3	26.2	21.4	21.6	(28)
(29)		2%	WB	13.6	16.1	16.7	20.4	23.0	25.0	26.8	26.2	25.2	23.7	18.5	17.6	(29)
(30)			MCDWB	15.0	18.5	19.5	23.5	26.6	29.0	30.4	29.4	27.5	25.4	20.2	19.3	(30)
(31)		5%	WB	11.6	13.8	15.6	18.9	22.3	24.3	26.1	25.6	24.7	22.6	17.0	15.7	(31)
(32)			MCDWB	13.2	15.7	18.2	22.0	25.4	28.1	29.9	28.4	26.8	24.2	18.5	17.0	(32)
(33)		10%	WB	9.9	11.1	13.7	17.5	21.6	23.8	25.5	25.0	24.1	20.7	15.4	13.6	(33)
(34)			MCDWB	11.4	13.2	16.0	20.3	24.4	27.3	29.2	27.6	26.1	22.5	17.1	14.8	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	7.5	8.0	8.1	8.5	7.6	7.4	7.3	7.0	7.5	7.8	7.7	(35)	
(36)			MCDBR	10.1	11.4	11.1	11.0	9.1	8.8	8.4	7.9	7.9	7.8	8.7	9.5	(36)
(37)			MCWBR	9.6	8.5	7.2	5.6	4.7	3.8	3.6	3.4	3.8	4.4	6.4	7.7	(37)
(38)		5% WB	MCDBR	9.8	11.1	9.9	9.1	7.7	7.7	7.8	6.9	6.2	6.3	7.8	9.3	(38)
(39)			MCWBR	9.5	9.2	7.3	5.7	4.6	3.9	3.8	3.7	3.5	4.1	7.1	8.1	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.311	0.320	0.346	0.393	0.436	0.482	0.513	0.507	0.412	0.355	0.325	0.314	(40)	
(41)		taud	2.547	2.515	2.474	2.353	2.266	2.186	2.114	2.118	2.376	2.529	2.561	2.568	(41)	
(42)		Ebn at Noon	883	917	920	888	851	809	780	775	839	865	859	855	(42)	
(43)		Edh at Noon	77	89	102	122	135	147	156	152	112	87	75	71	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.27	3.06	4.13	5.26	5.95	6.43	6.15	5.56	4.53	3.53	2.58	2.02	(44)	
(45)		RadStd	0.19	0.32	0.33	0.36	0.53	0.43	0.29	0.32	0.37	0.37	0.21	0.16	(45)	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (61 neighbors)	+0.47	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+158	+103

Nomenclature: See separate page